Dynamic Planet SSSS 2021 test (Thenerdicat)

Answer Sheet

1. (1 point) Freshwater has the salinity of less than ___ ppm

   Answer: 500 ppm
2. (8 points) How has sediment transport been impacted by human activity? Explain.
   a. (5 points) Explain how humans have contributed to an excess amount of sediment in watersheds

   Deforestation, poorly managed construction sites, winter road sanding, erosion/exposed soil, overgrazing/bad agricultural practice all lead up to excess weathering/erosion of surface sediments. The sediment eventually washes up into rivers and other watersheds, leading to excess sediments in watersheds. Dams also alter the patterns of sediment transport, adding on to the problem.

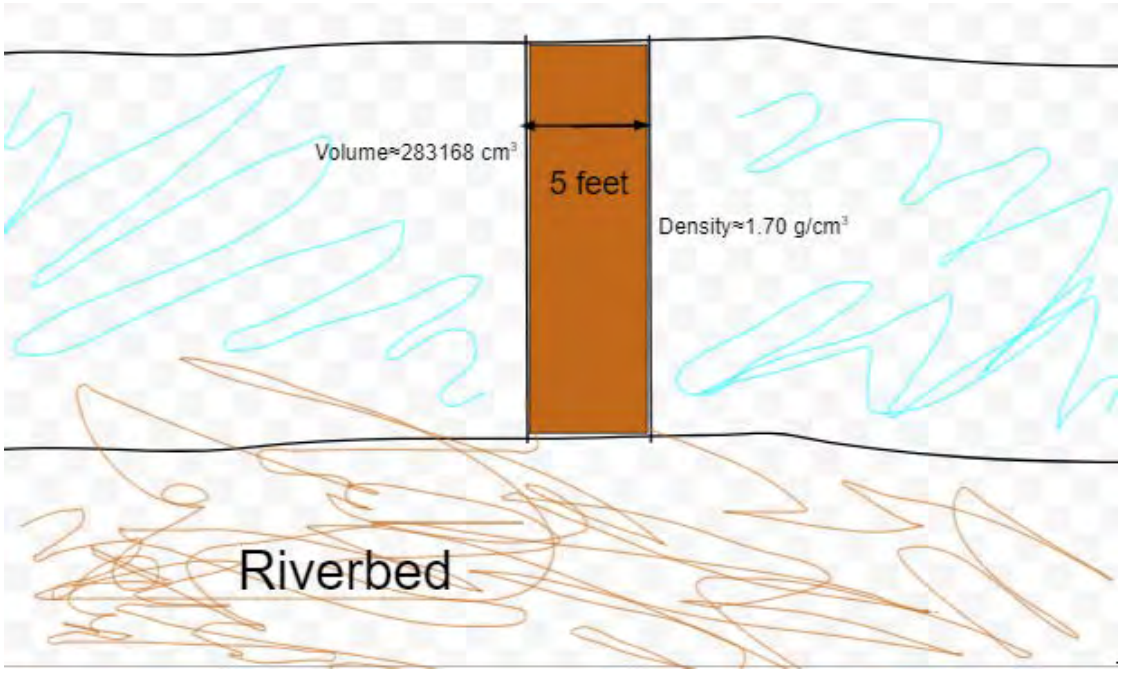
   b. (3 points) Explain how this affects water quality (things such as factors that influence water quality, etc.)

   Excess sediment leads to altered water depth and altered water flow, thus making watersheds more difficult to navigate. Water quality also goes down, such as increased turbidity, TDS (total dissolved solids), and TSS (total suspended solids). The extra sediment also disrupts the balance of watershed ecosystems, harming delicate indicator species and others alike.
3. (10 points) How do dams affect local watersheds? Explain.
   a. (5 points) Explain the things that change in watersheds when artificial dams are built

   Firstly, flow patterns are disrupted, thus affecting habitats as sediment transport patterns change. Adding on, water quality factors such as dissolved oxygen levels and temperature are changed. Cold, oxygen-deprived water is released from the reservoir bottom, affecting water quality. Sediments, logs, gravel, etc. can be trapped in dams, creating pools downstream and burying the riverbeds where fish spawn.

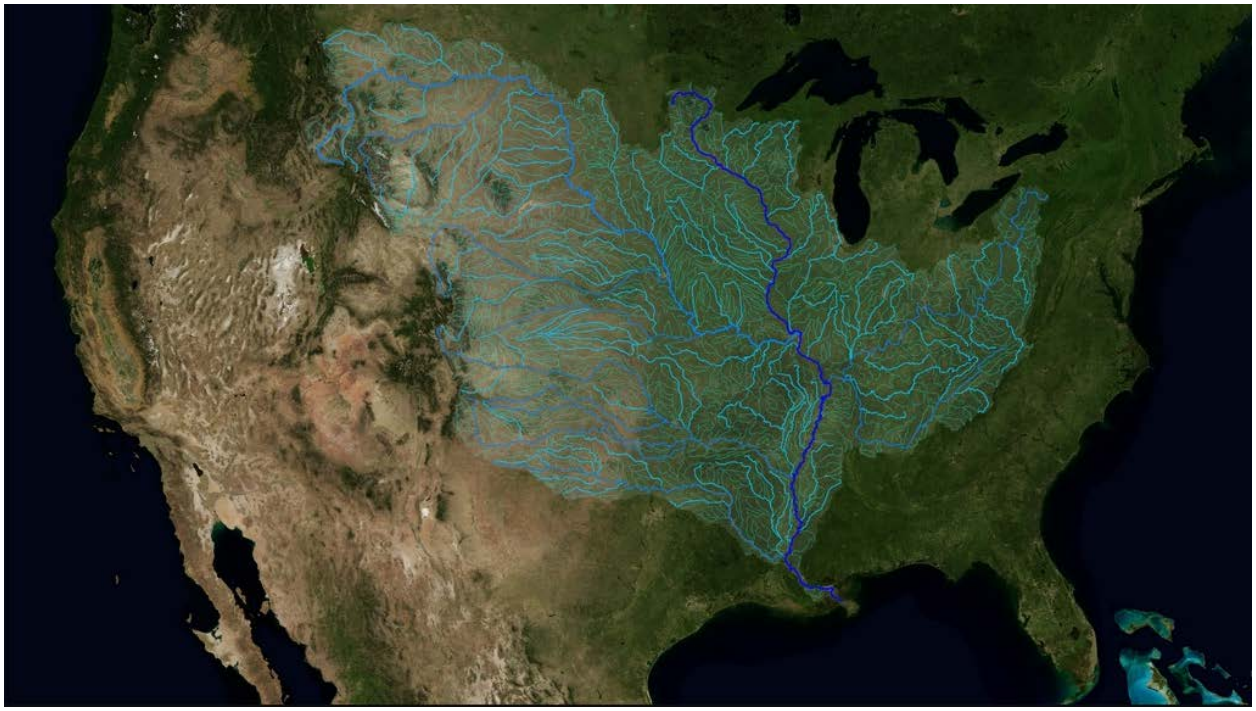
   b. (5 points) Explain how this affects local ecosystems

   Disrupted flow patterns affect migrating fish and disrupt the fragile ecosystem balance. Dissolved oxygen levels and temperatures are especially affected: surface water annual temperature variation increases while deep water annual temperature variation does the opposite; this is not good for sensitive species. Temperature also affects dissolved oxygen levels, thus leading to algal blooms and lower dissolved oxygen levels. Dams block rivers from migrating fish, disrupting the surrounding ecosystem once again.
4. (2 points) What is the average speed of sediment flow (per hour) based off the diagram below if the speed it took for the sediment to get transported was 10 hours? Express in grams/hour



Answer: 48138.56 grams/hour

5. (2 points) Below is a map of the Mississippi River drainage basin. What type of stream pattern does this follow?



Answer: Dendritic

6. (7 points) List the steps of watershed eutrophication. Explain how watersheds get an excess of nutrients (4 points) and then explain how watersheds "die out" from the excess of nutrients (3 points)

a. **(4 points) 1) Excess nutrients (ammonia/ammonium, nitrates, phosphates) and sediments are introduced to a watershed (mainly lakes), whether through human activity (i.e. an overabundance of fertilizer washing into a local lake) or naturally. 2) The introduction of abnormal levels of nutrients lead to algae rapidly blooming. This algal bloom; the water appears murky, covered in "scum" (the algae), and overall disgusting. Often, sediment washes in too, thus continuing the negative feedback loop of more algae growing.**
b. **(3 points) 3) Algae blocks sunlight from reaching the bottom of the affected area & uses up a lot of oxygen. 4) Eventually, the water becomes so depleted of oxygen organisms it becomes a "dead zone".**

7. (1 point) Lake Baikal's depths were first majorly explored in 1989 by a group of scientists from National Geographic Society & the Soviet Academy of Sciences. The life found at the bottom of the lake most closely resembles what type of ecosystem?
    A) Benthic life on the ocean floor (think benthic sediments)
    B) Life in hydrothermal vents (black and white smokers)
    C) Life in a coral reef (shallow waters, plenty of plant and animal life)
    D) Life in heavily saline lakes (Dead Sea as reference)

    Answer: B) Life in hydrothermal vents (black and white smokers)
8. (1 point) Which of these organisms <u>do not</u> live in a freshwater ecosystem?
    A) Mayfly
    B) Asian Carp
    C) Water hyacinth
    D) Mayfly
    E) Zooxanthellae
    F) Water lettuce

    Answer: E) zooxanthellae
9. (5 points) (2 points) Are watersheds like rivers and lakes part of the water table? (3 points) Why or why not?

    Answer: (2 points) Depends, but usually, yes. (3 points) It is usually part of the water table as the part where water table that naturally meets the surface is considered still a water table but it could be a watershed such as a stream, river, or lake. If we look at it a different way, it is not considered part of a water table as a watershed is a drainage basin where all precipitation drains to an endorheic basin, while a water table is the horizontal depth of the top of the aquifer. Thus, no and yes.
10. (3 points) What are the general steps of meander formation in streams? (2 points) How common are they? (1 point)

    Answer: Meanders are formed when water erodes sediment in the outer bank of a stream (2 points). They are mildly common especially in the middle & lower course of a bridge (1 point).
11. (6 points) What are the most common water pollutants in freshwater rivers worldwide?

    Answer: Phosphorus and nitrogen, as it comes from agricultural fertilizers and residential runoff. The nitrogen and phosphorus (especially) can lead to watershed eutrophication, eventually leading to "dead zones".
12. (2 points) Ox-bow lakes are formed from:

A) Meander scars
B) Floodplains
C) Meanders
D) Deltas

Answer: C), meanders

13. (2 points) Observe the photo below. Identify the type of wetland.



Answer: an estuarine delta

14. (2 points) Which drainage pattern is the most common?

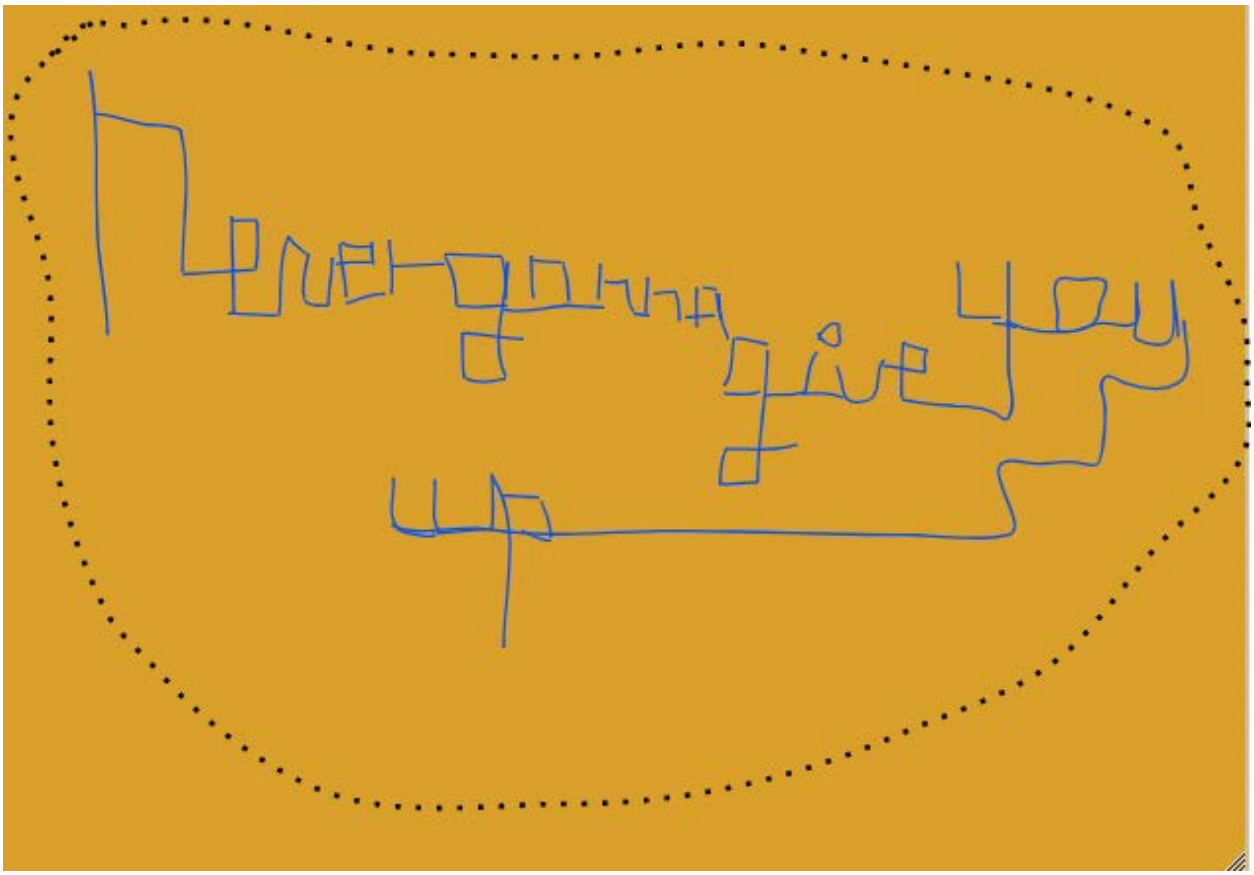
Answer: Dendritic drainage patterns are the most common drainage pattern.

15. (2 points) Which factor influences the quality of drinking water **the most**?

A) Temperature
B) pH
C) Dissolved oxygen
D) TDS

Answer: D), TDS

16. (2 points) What drainage pattern is the one pictured below? (The drainage pattern is in blue)



Answer: Rectangular drainage pattern.

17. (3 points) What is the main difference between a braided stream channel & an anastomose channel?

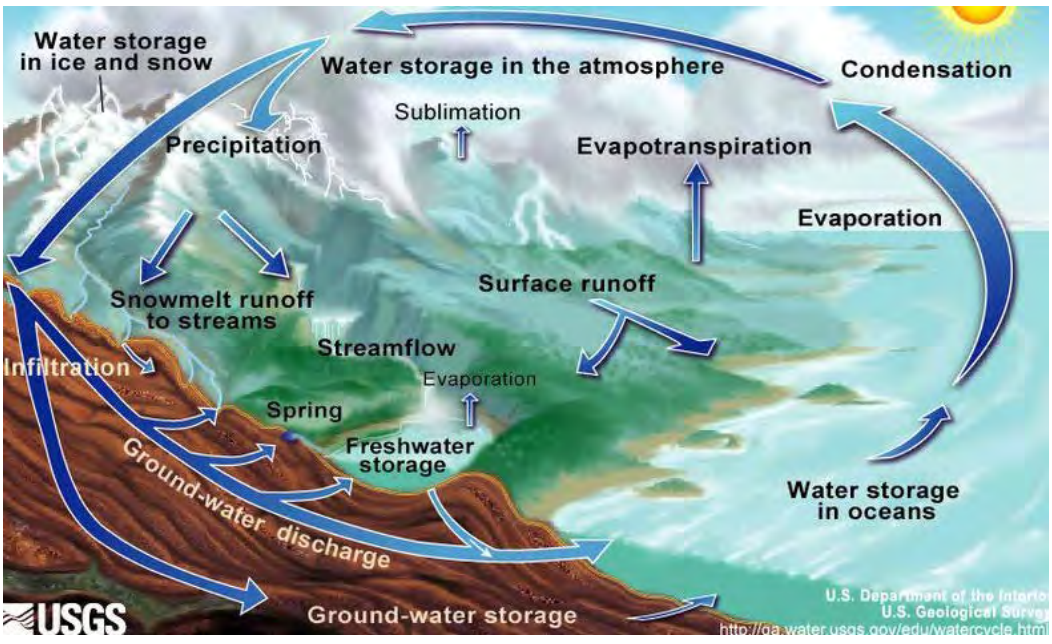
    Answer: Anastomose channels are basically braided stream channels, but they don't divide or reconnect as easily as braided stream channels.

18. (10 points) Explain the main steps for lake formation. (2 points) List out all 21 types of the lakes and explain how they form. (8 points)

    Answer: Each natural lake is formed when a depression called a *lake basin* gets filled in with water, or from an existing body of water. The way the lake basin forms varies. Cirque/tarn lakes are formed when the head of a glacier freezes and thaws, moving more rock material, forming a bowl-shaped basin, which is filled with the meltwater. Glacial lakes form when retreating glaciers carve the land and leave behind sediments that accumulate (moraines), which eventually get filled up with water (ex. Lake Agassiz). Rock hollow lakes are formed when glaciers scrape out depressions, which fill up into a lake. Volcanic lakes form as a remainder of volcanic activity. Often, the lake is a crater lake (caldera), which is a lake formed in a crater when a volcano/magma chamber collapses (ex. Crater Lake in Oregon formed from when Mount Mazama erupted ~7700 years ago and the magma chamber collapsed). It can be inferred that volcanic lakes can be formed from violent stratovolcanoes. Meteorite lakes are formed from the craters left behind from meteorite impact (ex. Lonar Lake in India). Solution lakes form when soluble underground rocks (i.e. limestone and salt) are dissolved into a basin from acidic running water; they are typically

round and permanent. Florida has a lot of solution lakes. Fluvial lakes are formed when sediment accumulates in a river bend which can form a separate lake or natural dam which forms a lake. It can be said that oxbow and landslide lakes can be counted as a type of fluvial lake, as well as lateral lakes. Oxbow lakes form when a meander's "neck" separates into an individual channel, forming an oxbow lake. Landslide lakes form when large amounts of fallen debris from a say, landslide (which is usually caused from rockfalls, mudflows, or iceslides), fall into a stream valley. This forms a natural dam, which forms a lake; these lakes are usually temporary as the dam can be eroded from the stream, unless the dam is especially large. Tectonic lakes are formed when tectonic activity (especially faulting). A depression called a tectonic basin or graben is formed usually as the result of an earthquake, which is filled with precipitation/groundwater. Arguably, the most famous tectonic lake is Lake Baikal in Siberia. Aeolian lakes are formed from wind action in usually arid areas; low-lying land in a desert with dunes lower than the groundwater level can collect water, which is another way which aeolian lakes form (ex. the lake group in the Tarim River, in Xinjiang, China). Freshwater lakes are lakes formed in any manner, fed from freshwater rivers with no sources flowing outwards. Organic/animal-made lakes are rather rare lakes formed from animal or plant activity, such as beavers damming up a river, or vegetation damming up a river forming a lake and coral lakes. Saline lakes form when a lake has water carrying salt & minerals with no way to escape as the lake has no outflow to other bodies of water (ex. Dead Sea). Barrier lakes (or damming) are formed when debris forms a dam and blocks a river up, turning it into a lake; they are rather temporary. Rift valley lakes are formed when a rift valley gets filled with precipitation, as the name suggests (ex. lakes along the East African Rift Valley). Karst lakes are formed when underground caves formed from water soluble rocks such as limestone, gypsum, and dolomite collapse; this process is called karstification (ex. Otjikoto Lake in Namibia). Wind deflation lakes are formed when wind erosion forms a deflated basin lake. They can be compared to aeolian lakes. Human-made/artificial lakes are made when humans construct an artificial lake for uses such as hydroelectric power, or say when human-made dams form a reservoir. A lake is permanent if the amount of evaporation is lower than the amount of precipitation, or if the lake is deep with a lot of water; a lake is temporary if it is in dry conditions or if the lake is formed when there is unusually high precipitation.

19. (2 points) Which cycle is being depicted below?



Answer: Hydrologic/water cycle

20. (2 points) Peat bogs are a ___ sink over thousands of years
    Answer: Carbon
21. (2 points) Glaciers and ice caps store __% of all water on Earth
    Answer: 2%
22. (6 points) How are waterfalls generally formed? Explain
    Answer: Waterfalls are usually formed when water flowing above harder rock erodes away at softer rock, downcutting it.
23. (6 points) How did this lake below form? Explain



    Answer: Lake Tanganyika was formed as a part of the western branch of the East African Rift, thus making it a rift lake. As how other rift lakes are formed, Lake Tanganyika was most likely formed from the rifting and subsidence of parts of the African plate ~9-12 million years ago.
24. (3 points) Why is sediment budget so difficult to calculate properly?
    Answer: sediment budget is difficult to calculate because it is difficult to calculate the exact amount of sediment added or subtracted from a certain area and all the sources and sinks of sediment must be taken into account.
25. (5 points) How has the Nile River's ability to provide water for daily human life been threatened? Hint: Human activities have caused the river to have poor water quality.
    Answer: The rapidly growing population needed a dam to provide enough water. The construction of a high dam at Aswan tampered with the river's delicate flow patterns, thus diminishing the seasonal flooding at the bank. The controlled outflow of water from the dam into Lake Nasser meant that it accumulated ~98% of the suspended sediment usually carried downstream. This is not good as agriculture required the rich sediment deposited when the river used to seasonally flood, leading to increased fertilizer use which leads to higher water pollution.
26. (2 points) Calculate the porosity of a cube of clay with the ratio of air to total volume as 4:16. Express as a percentage
    Answer: 25%
27. (2 points) What is considered a fast velocity of groundwater? (In centimeters/day)

Answer: 30.48 cm/day or greater (source: USGS)

28. (4 points) Order these materials from lowest to highest average porosity: salts, gravel, silt, clay, sand

    Answer: gravel, sand, silt, clay, and salts

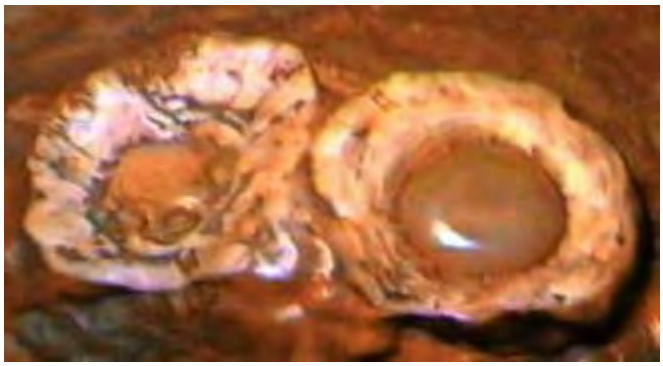
29. (6 points) What does Darcy's law state?

    Answer: Darcy's law states that the negative of permeability/dynamic viscosity of the fluid multiplied by the pressure drop is equal to the instantaneous flow rate of a fluid.

30. (1 point) _____ erodes from acid rainwater, forming karst formations. (Hint, usually in the form of calcite or aragonite.

    Answer: Limestone

31. (3 points) "Fried eggs" is a famous geological feature found in Luray Caverns. How did they form? (photo below)



    Answer: The "fried egg" formation was formed when workers accidentally snapped off two stalagmites, and tourists at Luray Caverns have eroded the stalagmites into the "fried egg" shape.

32. (5 points) Compare and contrast stalagmites and stalactites

    Answer: Both stalagmites & stalactites are formed when calcite salts precipitates from groundwater on cave walls; stalagmites precipitate from the floor up, while stalactites precipitate from the roof of the cave and up, almost like icicles.

33. (2 points) Find the vertical hydraulic gradient of a stream when the head difference taken from Point A to Point B is 13 meters, and the elevation difference is 7 meters. Round to the nearest thousandth (no fractions).

    Answer: 1.857 meters

34. (3 points) Is clay a common material found in calderas? Explain why or not.

    Answer: Yes, as after the volcano erupts and the depression is formed, the lake forms and clay called hectorite can accumulate.

35. (4 points) What is the process that water takes when it becomes surface runoff?

    Anwer: The water flows back into watersheds, is absorbed into the ground as groundwater, and erodes Earth's surfaces as it flows back into watersheds.

36. (4 points) Explain how karst formations form.

    Answer: Karst formations are formed when soft, water-soluble bedrock such as limestone, dolomite, gypsum, & marble gets dissolved by running water, forming distinct features such as sinkholes, springs, and streams.

37. (3 points) A pH of ____ or lower generally indicates poor water quality

Answer: 6.5

38. (4 points) How do sinkholes generally form?
    Answer: Sinkholes can be from either the erosion or removal of soluble bedrock (i.e. limestone) from running water, collapse of a cave roof, lowering of a water table, or overwithdrawal of groundwater.
39. (5 points) Why are geological features such as aquifers, aquitards, the water table, etc. important? What role do they have to play in ecosystems?
    Answer:
40. (3 points) What type of watershed is pictured below? How does it form and where is it generally found? (Be specific)



    Answer: The photo shows a tidal marsh. Tidal marshes form in areas that are sheltered from waves, and they drain and flood based off the tidal pattern of the adjacent watershed (i.e. rivers, estuaries, oceans). They are found along rivers, coasts, or seas, and can be freshwater or saline.
41. (5 points) In what ways is groundwater filtered? Why is it generally cleaner than surface water? How much cleaner?
    Answer: Groundwater is naturally filtered as it passes through layers of material and porous soil--groundwater is generally cleaner than surface water since surface water has chemicals, pollutants, organisms, and other materials that aren't filtered out like groundwater. Groundwater is usually safe to drink unlike surface water.
42. (5 points) Is clay's porosity higher than silt? Why or why not? Explain.
    Answer: No (generally), as the particles of silt are larger than clay, thus meaning silt has more pore space than clay, on average.

43. (3 points) How much arsenic in drinking water is safe? What is a "safe", government-issued level of arsenic in drinking water? (Hint, EPA)
    Answer: 0.010 mg/L or 10 ppb
44. (6 points) Why is salt in seawater bad for drinking? What is a safe level of dissolved salt in drinking water?
    Answer: The salt levels in seawater is way too high for the human body as human kidneys cannot get rid of the excess salt, plus seawater is dehydrating; there is no overarching issued "safe" level of dissolved salt in drinking water, USGS says to not exceed levels above 500 mg/L to avoid the salty taste.
45. (2 points) Why are seasonal lakes named ephemeral lakes?
    Answer: Seasonal lakes are named ephemeral lakes since "ephemeral" means "lasting temporarily", and seasonal lakes only hold water for short periods, thus ephemeral describes seasonal lakes well.
46. (2 points) What color of Secchi disks are used to measure freshwater turbidity?
    Answer: A black-and-white Secchi disk is generally used to measure freshwater turbidity.
47. (6 points) How does xeriscaping help watersheds?
    Answer: Xeriscaping, especially with native plants, reduces the amount of fertilizers and pesticides that get washed up into local watersheds, and, most importantly, conserves water.
48. (5 points) Why is Asian Carp so invasive to the Mississippi River and its tributaries?
    Answer: Asian Carp outcompetes native animals, disrupting the ecosystem. Plus, they can survive in lower quality water, killing off sensitive indicator organisms. Asian Carp also jump out of water to evade threats, which can injure people paddling in the Mississippi River and its tributaries. Lastly, they cost the Mississippi River area's fishing industry $7 billion annually by outcompeting native fish (those that are usually caught and sold).
49. (5 points) What role does the Chesapeake Bay watershed play in its ecosystem and the surrounding states?
    Answer: The Chesapeake Bay supplies a lot of seafood and recreational spots that many states rely on for economy, and the wetlands and watersheds shelter diverse communities of life and soften coastal storms.
50. (4 points) Why are cattails so important in preserving watersheds? Select all that apply
    A) They provide a habitat for organisms like birds to live among
    B) They soften heavy weather
    C) They filter runoff water, keeping watersheds clean
    D) They protect pond banks from erosion & reduce waves and winds hitting the shore
    Answer: All of the above but B)
51. (4 points) How are droughts typically predicted? Does the water table change when an area undergoes drought?
    Answer: The likelihood of droughts occurring are predicted by monitoring rainfall, river flow, and soil moisture. The water table changes, as the water table and aquifers rely on rainfall to replenish the levels of the water. The height of the water table drops as the rate water is pumped from the ground exceeds the rate the water can be replenished, thus the ground subsides and sinks.
52. (2 points) Explain the relationship between aquifers and groundwater
    Answer: An aquifer is a layer of underground permeable rock that holds groundwater.

53. (5 points) How does groundwater drilling affect the land and local water tables? Is this ethical for the environment?

    Answer: Groundwater drilling, or groundwater extraction causes land subsidence, lack of surface water, and lowered water quality in aquifers. If there is too much extraction of groundwater, the water table lowers thus leading to wells not being able to reach groundwater; watersheds like rivers that connect to groundwater will also have the water supply diminished. Thus, no, groundwater extraction is not ethical for the environment.

54. (2 points) An aquifer is usually found _____ the water table

    Answer: Below

55. (5 points) How does land subsidence affect groundwater and the water table?

    Answer: Land subsidence is sometimes due to the lowering of the water table or excess pumping of groundwater; the sediments that hold up the land with water now partially lowers due to the loss of groundwater, causing land to drop and changing the topographic gradient. Land subsidence also reduces aquifers' abilities to store water, most likely changing the water quality and altering underground sediments.

56. (6 points) What are some ways that humans impact groundwater quality?
    a. (3 points) Explain the actions of humans that impact groundwater
    b. (3 points) Explain the effects this has on groundwater quality

    Answer:

    a. **(3 points) Groundwater pumping for oil & gas extraction, groundwater extraction, contamination that seeps into the ground such as dumping toxic waste or landfill/waste disposal or agricultural contaminants, urban & industrial discharges all human activities that affects groundwater**
    b. **(3 points) Groundwater pumping contaminates groundwater and alters how water moves between a watershed, aquifer, and groundwater by intercepting groundwater flow that discharges into the surface-water body, or increasing the rate of water movement from the surface-watershed into an aquifer, leading to an aquifer holding too much water that discharges to surface watersheds that exacerbates erosion, altering aquatic life. Groundwater extraction leads to land subsidence and changes in topography; contaminants from things such as landfill waste can seep into the groundwater, again discharging to surrounding soil and watersheds, being toxic to life, decreasing ground- and surface- water quality; urban and industrial discharge might have similar effects to landfill waste.**

57. (4 points) How do pollutants such as arsenic and antimony get into drinking water?

    Answer: Arsenic gets into drinking water sources either naturally or artificially from industrial waste and agricultural pollution (certain fertilizers contain arsenic). Antimony gets into drinking water often from being dissolved in metal plumbing/fittings.

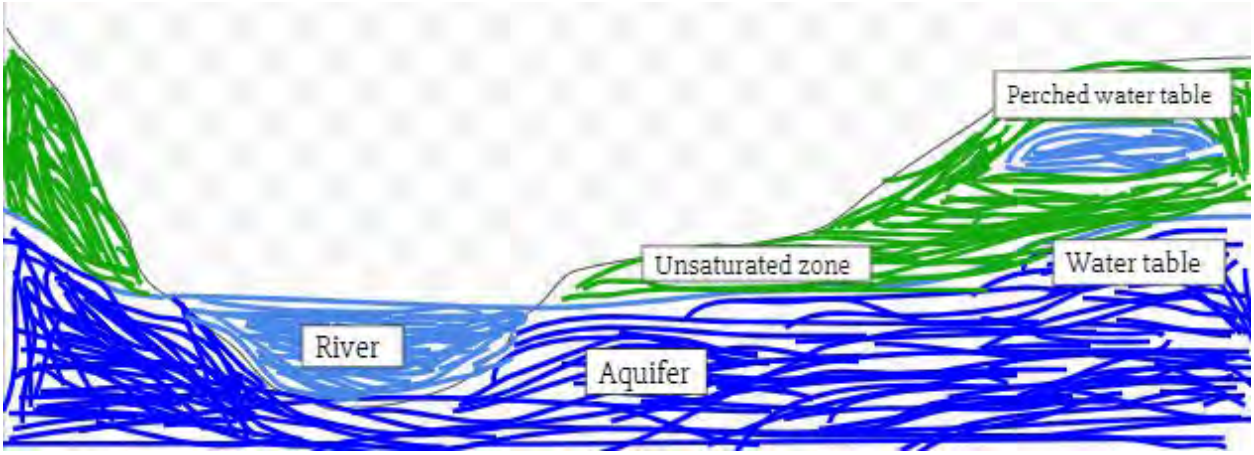
58. (2 points) On average, how many meters below ground is the unsaturated zone?

    Answer: the unsaturated zone is usually 0-100+ m underground, depending on climate and land type.

59. (3 points) How is the water balance equation an important part of hydrology?

Answer: The water balance equation is an important part of hydrology as it displays the system of hydrology as a whole using the conservation of mass.

60. (10 points) Label the diagram below



SSSS - Repaired practice edition - Answer key

Division B | Dynamic Planet | Repair appendix

This appendix contains authored replacements and supplied written alternatives. It does not reconstruct unpublished historical material. For the listed item IDs, use this appendix instead of the archived wording and key.

ANSWER KEY - Review only after submitting.

repair-case. (2 points) Authored written substitute for unavailable source material: Every lake mixes twice per year?

Unit: Lakes, wetlands, and freshwater storage.
Worked-case data: A storage feature lowers peak outflow but releases the same event volume later. Was water destroyed?

A. No

B. Always

C. Only if deep

D. Only if freshwater

Correct: A